

**Exponential Expressions**Question Bank - Additional 3.03 **ANSWER KEY**

Name

Date

Period

*Show your work. Circle your final answer.*

1. Two students evaluated  $\left(\frac{3^5}{3^2}\right)^2$ . Rylan first wrote  $\frac{3^5}{3^2} = 3^3$ , then  $(3^3)^2 = 3^6 = 729$ . Liam first wrote  $\frac{3^{10}}{3^4}$ , then  $3^6 = 729$ .

a. Name the law Rylan used first, and the law he used second.

***quotient of powers, then power of a power***

b. Name the law Liam used first, and the law he used second.

***power of a quotient (with power of a power on top and bottom), then quotient of powers***

c. List the laws needed to evaluate  $(4^3 \cdot 4^0)^2$ , then evaluate it.

***zero exponent (or product of powers), then power of a power;  $4^6 = 4,096$***

For questions 2 through 6, use the laws of exponents to evaluate. Show each law you use.

2.  $\left(\frac{2 \cdot 1^4}{2 \cdot 1^2}\right)^1$

***$2 \cdot 1^2 = 4,41$***

3.  $\left[\left(\frac{3}{4}\right)^3 \cdot \left(\frac{3}{4}\right)^0\right]^2$

***$\left(\frac{3}{4}\right)^6 = \frac{729}{4096}$***

*Math Nation's key prints 729/4,046 for this item;  $4^6 = 4,096$ .*

4.  $\frac{(4^2)^7}{4^{14}}$

***$\frac{4^{14}}{4^{14}} = 4^0 = 1$***

5.  $\frac{3.3}{4 \cdot 1^0} \cdot 3.3^2$

***$3.3 \cdot 3.3^2 = 3.3^3 = 35.937$***

6.  $\frac{(-6)^2 \cdot (-6)^1}{(-6)^0}$

***$(-6)^3 = -216$***

**Equivalent or not.**

7. Select ALL of the expressions that are equivalent to  $\frac{0.3^2}{3^2} \cdot 0.1^3$ .

- ☒ **A.**  $0.1^5$   
☐ B.  $0.3^0 \cdot 0.1^3$   
☒ **C.**  $0.1^2 \cdot 0.1^3$   
☒ **D.**  $\left(\frac{0.3}{3}\right)^2 \cdot 0.1^3$   
☐ E.  $0.1^6$   
☐ F.  $0.3^5$

**A, C and D**

$$0.1^2 \cdot 0.1^3 = 0.1^5 = 0.00001.$$

8. Which expression is equivalent to  $\left(\frac{7^6}{7^4}\right)^3$ ?

- ☒ **A.**  $7^6$   
☐ B.  $7^5$   
☐ C.  $7^{30}$   
☐ D.  $7^2$

**A**

**Working backwards.**

9. Find the value of  $n$  that makes each statement true.

a.  $\left(\frac{3^n}{3^2}\right)^2 = 3^8$

**$n = 6$**

b.  $\frac{2^n \cdot 2^0}{2^4} = 8$

**$n = 7$**

10. Kim says that  $\frac{(-18)^3}{6^3}$  can be rewritten as  $\left(\frac{-18}{6}\right)^3$  before evaluating. Is she right? Name the law, and give the value.

**Yes — power of a quotient, read from right to left;  $(-3)^3 = -27$ .**